ENVS 193DS Final

Matthew Schoen

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Link to ENVS 193DS Final GitHub Repo: HERE

Set Up

```
library(janitor)
library(tidyverse)
library(here)
library(readxl)
library(flextable)
library(DHARMa)
library(MuMIn)
library(ggeffects)

# Reading in the nest data & storing it as an object
nest <- read.csv(here("data", "nest_data_final.csv"))

# Reading in pesronal data & storing it as an object
step_count <- read_csv(
  here("data", "MyData_HW3_Format.csv"))
```

Problem 1: Research Writing

a. Transparents Statistical Methods

Part 1:

In part 1, they more than likely used a logistic regression as the response variable is the probably of American Avocet microhabitat use while the predictor variable is the distance to the water's edge.

Part 2:

In part 2, the they more than likely used a chi-square test of independence as both bird species and habitat type are categorical variables and the analysis looks to examine how the distribution of bird species differs among habitat type.

b. Suggestions for rewriting

Part 1:

American Avocet tuse of micro-habtitats varies wiht distance from the water edge. This suggests that proximity to water influences where and how the American Avocet chooses to use habitats.

Part 2:

American Avocets, Forster's Terns, and Black-necked Stilts used habitat types differently. This suggests that each species has habitat preference differs within the San Francisco Bay estuary.

c. Further Analysis

To further the analysis, examining how the type of vegetation cover within a given area influences American Avoocet habitat use. This *continuous variable* could be measured as a percentage (%) of ground cover within the study area using quadrats set at a given dimension (ex: 1m x 1m, 3ft x 3ft) and could showcase how other factors affected by vegetation cover (food, nesting area, protection from predators) influences behavior.

Problem 2: Data Analysis

Goal: Answer the following questions

1) How do tree height and acacia ant species influence the probability of bird nest occurrence (of any species of birds)?
2) Does acacia ant species identity influence the effect of tree height on probability of nest occurrence differ depending on acacia ant species?

a. Clean & Wrangle

```
# Creating new clean object from nest
nests_clean <- nest |>
  # selecting only the necessary columns
  select("case_control", # selecting case_control
         "height_cm", # selecting bird_speces
         "ant_species") |>  # selecting ant_species
  # filtering ant_species for the specified species
  filter(ant_species %in% c("RRB", "AB", "BBR")) |>
```

```r
  # reordering ant species as specied
  mutate(ant_species =
         fct_relevel(ant_species,
           "AB",
           "RRB",
           "BBR")) |>
  # Creating new column with spelled out names
 mutate(ant_species,
        ant_species_name = case_when( # specifying specific names in the new column
    ant_species == "AB" ~ "Crematogaster sjostedti",
    ant_species == "RRB" ~ "Crematogaster mimosae",
    ant_species == "BBR" ~ "Crematogaster nigriceps"
  ))

# Displaying structure of nests_clean
str(nests_clean)
```

```
'data.frame':	270 obs. of  4 variables:
 $ case_control    : int  1 0 0 0 0 1 0 0 0 1 ...
 $ height_cm       : num  392 310 513 154 324 ...
 $ ant_species     : Factor w/ 3 levels "AB","RRB","BBR": 2 2 2 2 1 2 3 2 1 3 ...
 $ ant_species_name: chr  "Crematogaster mimosae" "Crematogaster mimosae" "Crematogaster mimo
```

```r
# Displaying 10 random rows within the data set
slice_sample(nests_clean, n = 10)
```

```
   case_control height_cm ant_species      ant_species_name
1             0     166.5         RRB Crematogaster mimosae
2             1     212.5         RRB Crematogaster mimosae
3             1     191.0         RRB Crematogaster mimosae
4             0     114.5         RRB Crematogaster mimosae
5             0     121.5         RRB Crematogaster mimosae
6             1     194.0         RRB Crematogaster mimosae
7             1     511.5         RRB Crematogaster mimosae
8             0     158.5         RRB Crematogaster mimosae
9             0     175.0         RRB Crematogaster mimosae
10            0     446.0         RRB Crematogaster mimosae
```

b. Response Variable

In this data set, "case_control" is made up of a binary system where 1 indicates a bird nest is present in a tree while 0 indicates no bird nests were found within a tree. This is important when assessing how/why birds choose specific trees for nesting.

c. Purpose of the Study

Tree Height:

Tree height is a continuous variable measured in centimeters. As tree height increases, the probability of nest occurrence increases as a taller tree provides greater protection from predators, visibility, and increases structural support to the nest itself (Found in the "Introduction" section).

Ant Species:

Ant species is a discrete categorical variable as it contains three distinct "categories" which in this case are the different species of ants: *Crematogaster sjostedti*, *Crematoogaster mimosae*, and *Crematoogaster nigriceps*. The species of ant can influence nesting probability as different species of ants vary in their overall "aggressiveness" toward other organsisms which in turn alters tree structure and can affect the overall suitability of a tree for nesting. (Found in Introduction & Methods).

d. Table of Models

Model number	Tree Height	Ant Species	Interaction	Predictor list
1	X	X	X	All predictors (Tree height and ant species) with an interaction
2	X	X		Tree Height and Ant Species (No Interaction)

Figure 1: Table for Models

e. Run the Models

f. Select the Best Model

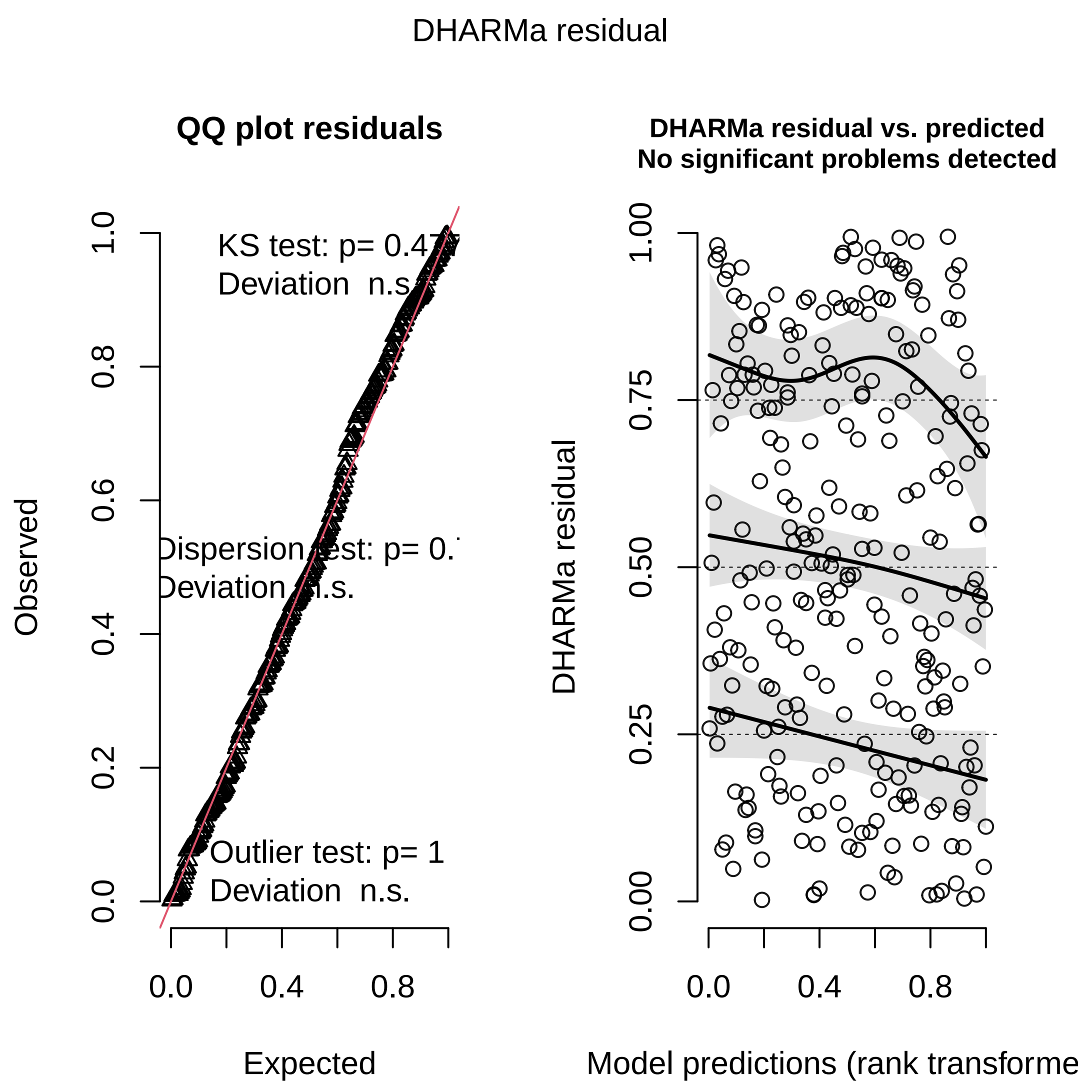
```
# Running an AIC
AIC(
  model1,
  model2)
```

```
       df      AIC
model1  6 237.8042
model2  4 258.7430
```

The best model as determined by Akaike's Informationi Criterion (AIC) included both predictors, tree height (cm) and ant species, and the interaction between both of them (AIC = 237.8). The results suggest that a relationship exists between tree height and the probability of nest occurrence based off the species of acacia ant.

g. Check the Diagnostics

```
# Plotting Residuals
plot(simulateResiduals(model1))
```



h. Visualize the Model Predictions

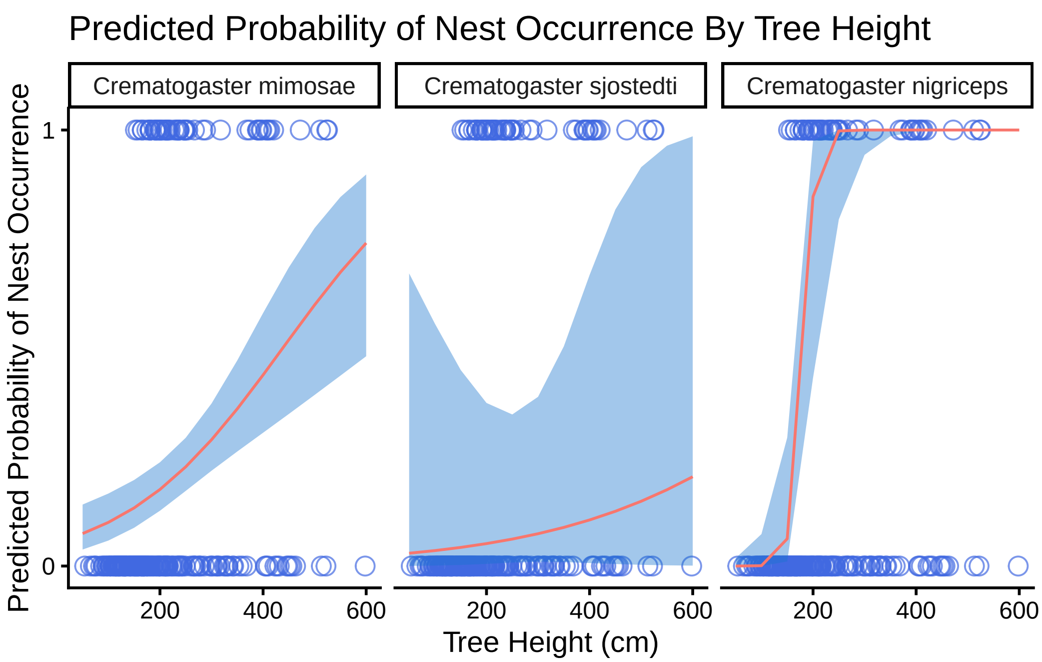
```
preds <- ggpredict(
  model1,
  terms = c("height_cm",
            "ant_species_name"))
```

```
Some of the focal terms are of type `character`. This may lead to
```

```
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `ant_species_name`
```

```
Data were 'prettified'. Consider using `terms="height_cm [all]"` to get
  smooth plots.
```

```
# ggplot as base layer
ggplot(data = nests_clean,
       aes(x = height_cm,
           y = case_control)) +
  geom_point(size = 3,
             alpha = 0.7,
             shape = 21,
             color = "royalblue") +
  geom_ribbon(data = preds,
              aes( x = x,
                   ymin = conf.low,
                   ymax = conf.high,
                   group = group),
                  fill = "dodgerblue3",
                   alpha = 0.4,
              inherit.aes = FALSE) +
  geom_line(data = preds,
            aes(x = x,
                y = predicted,
                color = "dodgerblue"),
            inherit.aes = FALSE) +
  facet_wrap( ~ group) +
  scale_y_continuous(limits = c(0, 1),
                     breaks = c(0, 1)) +
  labs(
    x = "Tree Height (cm)",
    y = "Predicted Probability of Nest Occurrence",
    title = "Predicted Probability of Nest Occurrence By Tree Height") +
  # removing excess visuals with updated theme
  theme_classic() +
  # removing the legend
  theme(legend.position = "none")
```



i. Write A Figure Caption

Figure 3

j. Calculate Model Predictions

k. Interpret Your Results

Problem 3: Affective and Exploratory Visualizations

a. Comparing Visualizations

How are the visualizations different from each other in the way you have represented your data?

The exploratory visualizations span between a bar chart comparing mean activity level per day of the week and a jitter plot examining if there is a direct relationship between time slept and activity level while the affective visualization is a bar chart that combines my predictor and response variables as well as my other predictory variables. The exploratory visualizations look to examine direct relationships between one predictor variable and the response while the the affective visualization serves as a holistic view of how general physical exertion and recovery habits influence overall daily activity level. It should also be noted that the affective visualization relies heavily on *color* while the exploratory visualizations use basic numerical-based figures.

What similarities do you see between all your visualizations?

The main takeaway from all of the visualizations is that there isn't just *one* deciding factor that directly influences my daily activity level. This is illustrated in contrasting the exploratory visualizations (comparing one predictor to the response variable) with the affective visualization: the exploratory pieces showcase *some* form of a relationship but not a necessarily strong one while the affective piece shows that the additional variables I measured also have a drastic impact on overall activity level. In other words, all the visualizations showcase that *multiple factors contribute to my overall daily activity level*.

What patterns (e.g. differences in means/counts/proportions/medians, trends through time, relationships between variables) do you see in each visualization? Are these different between visualizations? If so, why? If not, why not?

In first exploratory visualization (comparing daily activity per day of the week), during the week, mean daily step count is generally higher with Tuesday and Thursday exhibiting the highest overall steps taken while the weekend days (Saturday & Sunday) have the exhibit the lowest steps taken. In the second visualization (comparing time slept and activity level), there seems to be a somewhat negative correlation between time slept and activity level where activity level decreases as time slept increases. In comparing all three visualizations, the findings from the exploratory visualizations match that of the affective visualization where my main recovery days (weekends) have lower overall activity level.

What kinds of feedback did you get during week 9 in workshop or from the instructors? How did you implement or try those suggestions? If you tried and kept those suggestions, explain how and why; if not, explain why not.

During week 9, I didn't get much feedback from my peers other than the fact that they liked where I was going with my affective visualization. They also thought that the variables I used to collect information my "study" were well thought out and supported finding an answer to my central question. The main feedback I got from instructors were regarding my exploratory visualizations and this was mainly just to clean up the plots which I did!

b. Designing An Analysis

Question: Is there a statistical relationship between time slept and activity level (measured in daily steps taken)?

Analysis: To understand whether there is a statistical relationship between total time slept (minutes) and activity level (measured in daily steps taken), I will run a correlation test. This is appropriate as based off my initial exploratory visualization, it seems that there is *some* form of a negative linear relationship between the total time slept and activity level the next day. However, because the relationship isn't strictly obvious, I want to assess how correlated these variables are. Understanding this will aid in how I prioritize sleep in my every day life.

Variables Assessed

Predictor Variable: Total Time Slept On A Nightly Basis (Minutes)

Type: Numerical, Continuous

Response Variable: Total Steps Taken Daily

Type: Numerial, Discrete

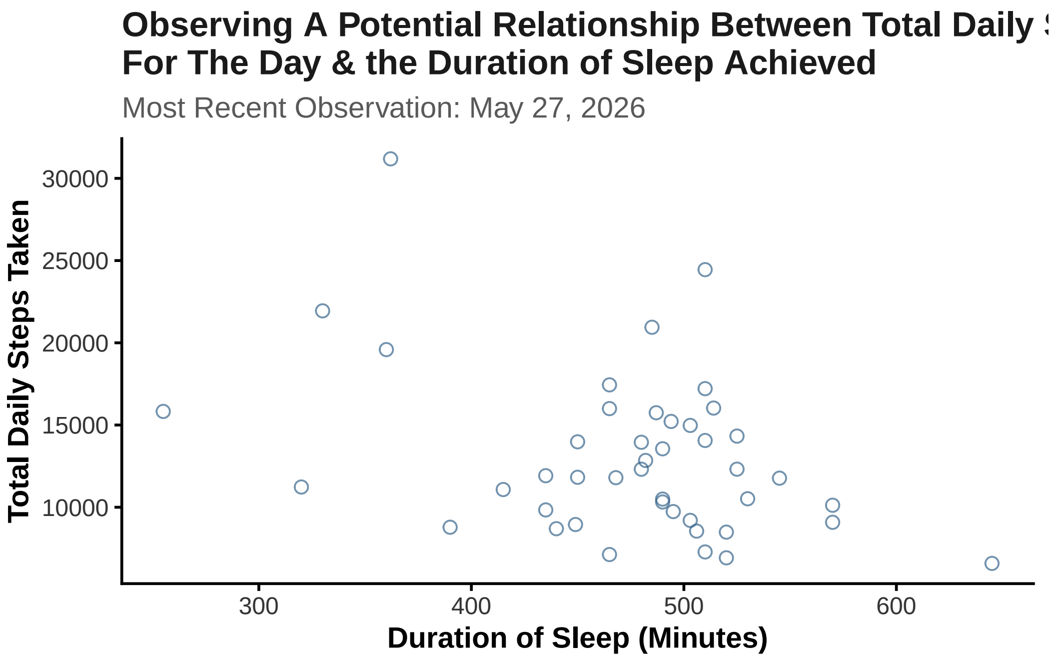
c. Check Any Assumptions and Run Your Analysis

Checking Assumptions

```
# cleaning my data before creating ggplot
my_data_clean <- step_count |>
  # cleaning names
  clean_names() |>
  # removing any NA values
  filter(!is.na(day_of_the_week))
```

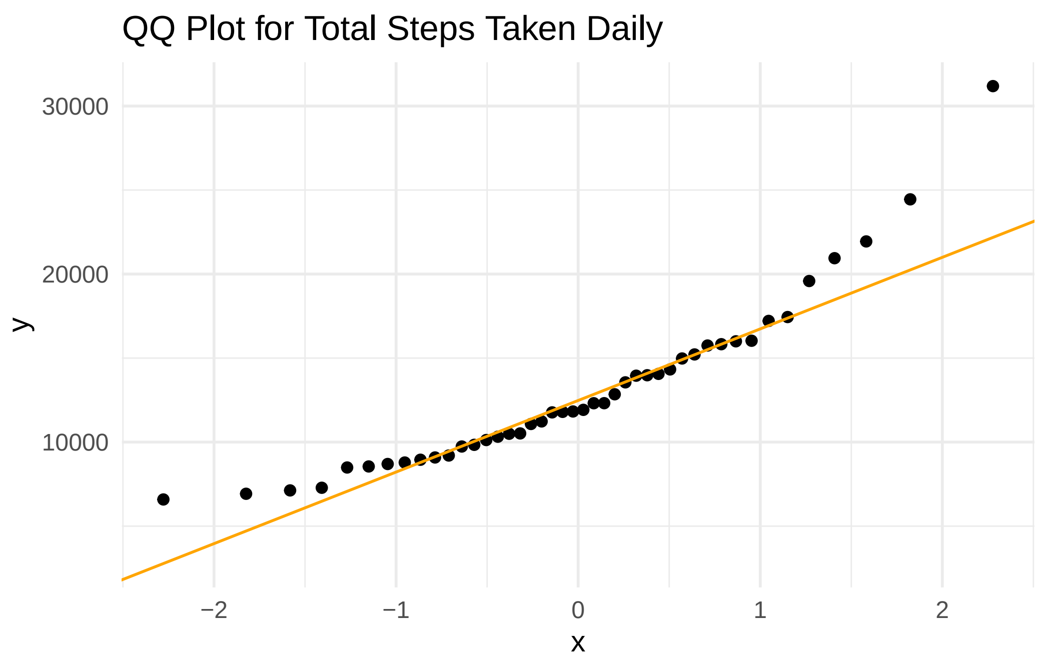
```
#creating ggplot base layer
ggplot(data = my_data_clean,
       mapping = aes(x = duration_of_sleep_minutes,
                     y = total_steps_taken_for_the_day)) +
  # Adding a QQ plot
  geom_point(color = "steelblue4",
             size = 2,
             alpha = 0.7,
             shape = 21) +
  # Adding custom labels for the x and y axis
  labs(x = "Duration of Sleep (Minutes)",
       y = "Total Daily Steps Taken",
       title = "Observing A Potential Relationship Between Total Daily Steps Taken
For The Day & the Duration of Sleep Achieved",
      subtitle = "Most Recent Observation: May 27, 2026") +
  # Updating the theme to remove grid lines
  theme_classic() +
  # Customizing the theme elements
```

```
  theme(plot.title = element_text(face = "bold", # bolding the title
                                  color = "gray10"), #making the title gray
        plot.subtitle = element_text(color = "gray35"), # choosing color
        axis.title = element_text(face = "bold"), # bolding the axes
        axis.text = element_text(color = "gray20")) # axes color
```

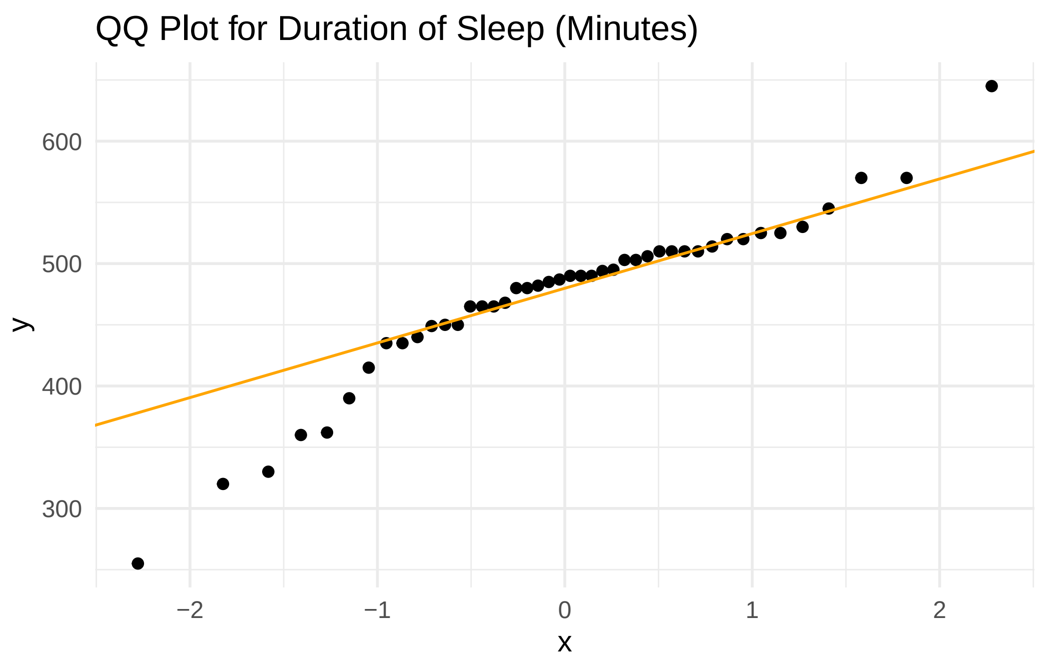


Note: There seems to be some type of negative relationship between the two but need to ensure the normality of the data before running the correlation test.

```
# ggplot as base layer
  ggplot(data = my_data_clean,
         mapping = aes(sample = total_steps_taken_for_the_day)) + # specifying use of total s
  # first layer is QQ plot
  geom_qq() +
  # second layer is QQ line
  geom_qq_line(color = "orange") + # making line red
  # updating theme
  theme_minimal() +
  # updating title
  labs(title = "QQ Plot for Total Steps Taken Daily")
```




```
# ggplot as the base layer
   ggplot(data = my_data_clean,
    mapping = aes(sample = duration_of_sleep_minutes)) +
  # specifying use of sleep duration
  # first layer is the qq plot
  geom_qq() +
  # second layer is the qq plot line
  geom_qq_line(color = "orange") + # making line red
  # updating theme
  theme_minimal() +
  # updating title
  labs(title = "QQ Plot for Duration of Sleep (Minutes)")
```



Note: The data for both Total Daily Steps Taken and Duration of Sleep (minutes) appear *relatively normal*. However, the scatter plot directly comparing the variables showcases there could be other factors influencing the relationship. As a result, I believe running both a Pearson's & Spearman-rank correlation test and then comparing the results of both would be beneficial while enhancing the rigor of my analysis.

Running Proposed Analysis

```
# Conducting a Pearson Correlation test
cor.test(my_data_clean$total_steps_taken_for_the_day,
         my_data_clean$duration_of_sleep_minutes,
         method = "pearson")
```

```
    Pearson's product-moment correlation

data:  my_data_clean$total_steps_taken_for_the_day and my_data_clean$duration_of_sleep_minute
t = -2.6228, df = 42, p-value = 0.0121
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.60468177 -0.08807612
```

```
sample estimates:
       cor
-0.3751471
```

```
# Conducting a Spearman-rank correlation test
cor.test(my_data_clean$total_steps_taken_for_the_day,
         my_data_clean$duration_of_sleep_minutes,
         method = "spearman")
```

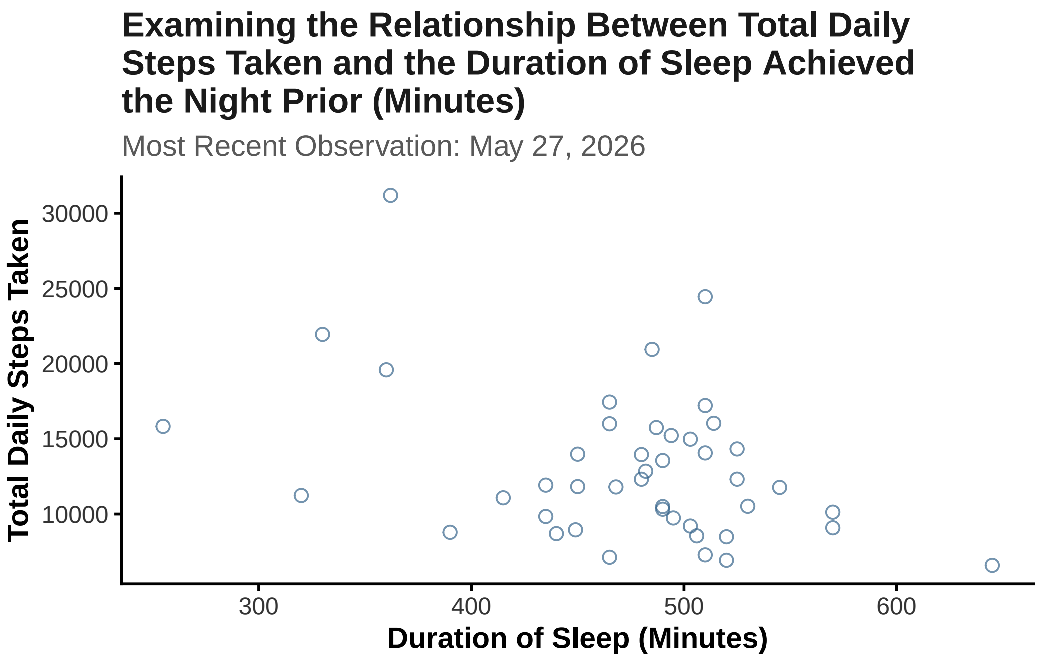
```
Warning in cor.test.default(my_data_clean$total_steps_taken_for_the_day, :
Cannot compute exact p-value with ties
```

```

    Spearman's rank correlation rho

data:  my_data_clean$total_steps_taken_for_the_day and my_data_clean$duration_of_sleep_minute
S = 17801, p-value = 0.0955
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
-0.2544878
```

d. Create a Visualization

```
#creating ggplot base layer
ggplot(data = my_data_clean,
       mapping = aes(x = duration_of_sleep_minutes,
                     y = total_steps_taken_for_the_day)) +
  # Adding a QQ plot
  geom_point(color = "steelblue4",
             size = 2,
             alpha = 0.7,
             shape = 21) +
  # Adding custom labels for the x and y axis
  labs(x = "Duration of Sleep (Minutes)",
       y = "Total Daily Steps Taken",
       title = "Examining the Relationship Between Total Daily
Steps Taken and the Duration of Sleep Achieved
the Night Prior (Minutes)",
      subtitle = "Most Recent Observation: May 27, 2026") +
```

```
  # Updating the theme to remove grid lines
  theme_classic() +
  # Customizing the theme elements
  theme(plot.title = element_text(face = "bold", # bolding the title
                                  color = "gray10"), #making the title gray
        plot.subtitle = element_text(color = "gray35"), # choosing color
        axis.title = element_text(face = "bold"), # bolding the axes
        axis.text = element_text(color = "gray20")) # axes color
```



e. Write a Caption

Figure Caption:

The figure depicts the relationship between sleep duration (minutes) shown on the x-axis and total daily step count across all observations collected during the study period shown on the y-axis. The points represent the individual observations recorded during the study period.

f. Write About Your Results

The Pearson correlation test identified a *weak negative relationship* between total daily steps and duration of sleep (Pearson's product-moment correlation, Pearson's r = -0.375, t(42) =

-2.62, p = 0.012, α = 0.05, CI[-0.605, -0.088]). Because the p-value was less than α = 0.05, the null hypothesis of no linear relationship was rejected. However, the Spearman-rank correlation test *did not* identify a statistically significant monotonic relationship between the two variables (Spearman rank correlation test: ρ = -0.254, S = 17801, p = 0.096). Since the Spearman p-value was greater than α = 0.05, the null hypothesis was not rejected for the rank-based test. Given the conflicting results between the two tests, it can be said that while there *may* be somewhat of a relationship between total time slept and total daily steps taken, other variables (such as the additional variables measured) more than likely influence overal step count.

g. Sharing Your Affective Visualization

Completed during week 10 in workshop!